

SAT Math Mastery · Trick Pack 1I

SAT Trick Pack 1 — Part I

Time 75 minutes: 55 min solving + 20 min self-check and error log

Topics rewriting exponentials · parallel lines · inscribed right triangles · factor pairs · mixtures · rates and units · conjugate pairs · tangency · sequential transfers · chords · solids in cross-section · circular segments · data transformations

Name / Date _____

1 · SKILLS & IDEAS

- S1. Rewrite a shifted exponential with one base and read the initial value; a fractional base with a negative exponent grows.
- S2. Recognise parallel, coincident and crossing lines from proportional coefficients.
- S3. Spot a diameter among the sides of an inscribed triangle and use the right angle opposite it.
- S4. Optimise a parameter over factor pairs, weighting each pair by its coefficients.
- S5. Track one component through successive mixing stages and write one equation for the unknown.
- S6. Combine a total-area equation and a total-volume equation using rates in the correct direction.
- S7. Add fractions in conjugate pairs $(1 \pm a, 1 \pm a^2, \dots)$ and then invert.
- S8. Use a zero discriminant (or centre-to-line distance) for tangency, and report the quantity actually asked for.
- S9. Follow sequential fractional transfers between two groups with one variable.
- S10. Convert units one factor at a time; time units are squared in an acceleration.
- S11. Compute a chord from the centre-to-line distance and the radius.
- S12. Reduce a solid to its axial cross-section; $\text{inradius} = \text{area}/\text{semiperimeter}$.
- S13. Segment area = sector – central triangle; special right triangles give the central angle.
- S14. Decide what a shift or an appended value does to the mean, median and standard deviation.

2 · REQUIRED FACTS

- $(\frac{1}{b})^{-u} = b^u$ and $(b^p)^q = b^{pq}$; the value at $x = 0$ of ab^x is a .
- Two lines with proportional x - and y -coefficients are parallel if the constants are not in the same proportion, and identical if they are.
- An angle inscribed in a semicircle is a right angle, so a chord equal to the diameter is the hypotenuse of a right triangle.
- In a mixture, only the *component* adds: $\text{amount} = \text{percent} \times \text{mass of that batch}$.
- A line is tangent to a circle exactly when the distance from the centre to the line equals the radius, equivalently when the substituted quadratic has discriminant 0.
- Rates may be given as “amount per area” or “area per amount”; these are reciprocals.
- Squaring a time-unit factor: $1 \text{ min}^2 = 3600 \text{ s}^2$.
- Adding a constant to every value shifts the mean and median by that constant and leaves the standard deviation unchanged; appending the mean leaves the mean and the sum of squared deviations unchanged.

3 · REQUIRED FORMULAS

$$\frac{1}{1-a} + \frac{1}{1+a} = \frac{2}{1-a^2}, \quad \frac{2}{1-a^2} + \frac{2}{1+a^2} = \frac{4}{1-a^4}, \quad \ell = 2\sqrt{r^2 - d^2}, \quad d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

$$r_{\text{in}} = \frac{\text{Area}}{s}, \quad A_{\text{segment}} = \frac{\theta}{360^\circ} \pi r^2 - \frac{1}{2} r^2 \sin \theta, \quad V_{\text{sphere}} = \frac{4}{3} \pi r^3, \quad V_{\text{cone}} = \frac{1}{3} \pi R^2 h$$

4 · MINI THEORY SHEET

One base, then read. Write $(\frac{1}{8})^{(2-x)/3} = 2^{-(2-x)} = 2^{x-2}$ and split off the constant: $48 \cdot 2^{x-2} = 12 \cdot 2^x$. The coefficient is the value at $x = 0$.

Mixtures. For each stage write (percent)(mass) for the component only; pure additions contribute 100% of their mass. One equation: component after = percent after $\times$ mass after.

Tangency. Substitute the line into the circle; setting the discriminant to 0 gives an equation in the slope. Check whether the question asks for m or m^2 .

Cross-sections. A sphere inscribed in a cone is a circle inscribed in the axial triangle; its radius is area over semiperimeter.

Data. Work with the *sum of squared deviations*: a shift leaves it alone and appending the mean adds 0 to it, while the count grows.

Speed shortcuts.

Situation	Shortcut
$k \cdot b^{(c-x)/n}$ with $0 < b < 1$	rewrite with base $1/b$; this is growth, not decay; $f(0)$ is the coefficient
fractions with proportional left sides	same line or parallel: compare the constants after scaling
side of an inscribed triangle equals $2r$	right angle opposite it; Pythagorean triple for the missing side
$(pu + a)(qu + b)$ with fixed ab	$k = pb + qa$: pair the larger factor of ab with the larger coefficient
add pure component, then another alloy	one equation: all tin after = percent $\times$ total mass after
coverage rates in different forms	write litres per square metre for each; two equations: area and volume
$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \dots$	pair conjugates; the sum collapses step by step
line meets circle once	distance from centre to line = r ; solve for m^2 directly
fraction of A leaves, fraction of B returns	write A and B after each move in terms of x
acceleration in a new time unit	multiply by (time-unit factor) ²
chord from line and circle	d first, then $2\sqrt{r^2 - d^2}$; no intersection points needed
sphere in cone	slice through the axis; $r = \text{Area}/s$
chord and radius form a 30° - 30° - 120° pattern	read the central angle, then sector – triangle
shift then append the mean	mean shifts, SD falls, median must be re-checked

5 · GUIDED EXAMPLES

Example 1 — Pair the conjugates, then invert

For real t with $t^2 \neq 1$, $\frac{1}{1-t} + \frac{1}{1+t} = 5$. Find t^2 .

$$\frac{1}{1-t} + \frac{1}{1+t} = \frac{2}{1-t^2} = 5 \Rightarrow 1-t^2 = \frac{2}{5} \Rightarrow t^2 = \frac{3}{5}.$$

Example 2 — Chord from the distance, not the intersection points

Find the length of the chord cut from $x^2 + y^2 = 25$ by the line $x + y = 7$.

$$d = \frac{|0+0-7|}{\sqrt{2}}, \text{ so } d^2 = \frac{49}{2}. \text{ Chord} = 2\sqrt{25 - \frac{49}{2}} = 2\sqrt{\frac{1}{2}} = \sqrt{2}.$$

Example 3 — Inradius from area and semiperimeter

An isosceles triangle has sides 5, 5 and 6. Find the radius of its inscribed circle.

$$\text{Height} = \sqrt{25 - 9} = 4, \text{ area} = 12, \text{ semiperimeter} = 8. \quad r = \frac{12}{8} = \frac{3}{2}.$$

6 · PRACTICE SET — 15 PROBLEMS · ALL SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined for $x \geq 0$ by

$$f(x) = 48 \left(\frac{1}{8} \right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 192 \cdot \left(\frac{1}{2} \right)^x$

(B) $f(x) = 12 \cdot 2^x$

(C) $f(x) = 48 \cdot 2^x$

(D) $f(x) = 12 \cdot \left(\frac{1}{2} \right)^x$

2.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{x}{6} + \frac{y}{4} = 2$$

Which of the following equations, when paired with the equation above, forms a system of two linear equations in x and y that has no solution?

(A) $4x - 6y = 40$

(B) $4x + 6y = 48$

(C) $4x + 6y = 40$

(D) $6x + 4y = 48$

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

Triangle ABC is inscribed in a circle with circumference 52π . Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?

(A) $\frac{10}{13}$

(B) $\frac{24}{13}$

(C) $\frac{5}{12}$

(D) $\frac{5}{13}$

4.

SAT CHALLENGE • NO CALCULATOR NEEDED

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k ?

5.

SAT CHALLENGE • CALCULATOR

A metalworker melts x kilograms of an alloy that is 30% tin and mixes in 15 kilograms of pure tin. She then adds 25 kilograms of a second alloy that is 10% tin. The final alloy is 35% tin. What is the total mass, in kilograms, of the final alloy?

- (A) 125 (B) 70
(C) 85 (D) 110

6.

SAT CHALLENGE • CALCULATOR

A contractor finishes a 96-square-meter floor using two products. Primer P is applied in two coats, and each coat uses 125 milliliters per square meter. Sealant Q is applied in one coat and covers 5 square meters per liter. Every square meter receives exactly one of the two products, and the job used a total of 21.2 liters. How many square meters did the contractor cover with primer P?

7.

SAT CHALLENGE • NO CALCULATOR NEEDED

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8 ?

- (A) $\frac{5}{4}$ (B) $\frac{9}{5}$
 (C) $\frac{1}{5}$ (D) $\frac{4}{5}$

8.

SAT CHALLENGE · CALCULATOR

In the xy -plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2 ?

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

(A) 72

(B) 36

(C) 45

(D) 63

10.

SAT CHALLENGE · CALCULATOR

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $300 \cdot 2^{m/18}$

(B) $1200 \cdot 2^{m/36}$

(C) $900 \cdot 2^{m/18}$

(D) $1200 \cdot 2^{m/18}$

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q . What is the length of $\overline{PQ}$?

(A) 6

(B) $2\sqrt{21}$

(C) 8

(D) 3

13.

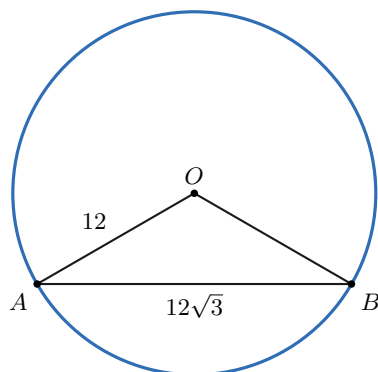
SAT CHALLENGE · NO CALCULATOR NEEDED

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB ?



(A) $24\pi - 36\sqrt{3}$

(B) $48\pi - 36\sqrt{3}$

(C) 48π

(D) $96\pi + 36\sqrt{3}$

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.

- I. The mean is 4 greater.
- II. The standard deviation is smaller.
- III. The median is 4 greater.

Which of the statements must be true?

- (A) I, II and III
- (B) I only
- (C) I and II only
- (D) II and III only
